Adaptation model of accommodation and vergence

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Both accommodation and vergence have been shown to exhibit adaptation after extended near viewing. Normally, when the stimulus to accommodation is removed, the accommodation system returns rapidly towards its tonic position. However, if the stimulus is removed after an extended focusing effort, the decay is much slower. A similar effect can be observed in the vergence system. After prolonged wearing of horizontal prisms, blockage of one eye results in a much slower decay of the vergence output towards its tonic value. No previous models have been shown to simulate quantitatively these effects. An interactive dual-feedback model of accommodation and vergence was developed to simulate the adaptive behaviour found experimentally. The unique feature of the model is that the output of each controller drives a dynamic adaptive component whose output governs the time constant of the controller. The model was able to simulate the rapid and slow decays following short and long viewing intervals in each of the accommodative and vergence systems. It also simulated adaptation during alternate binocular and monocular viewing under the accommodation closed-loop condition. Thus, this model can serve as the basis for detailed quantitative evaluation of adaptive behaviour in the accommodation and vergence systems.

Most biological systems exhibit the ability to adapt to changing conditions in the environment. Sensory adaptation, such as in the olfactory and auditory systems, serves to shift the operating level over a wide dynamic range. Motor adaptation, such as in muscles, serves to maintain accuracy of movements despite changes in external load. Adaptation is useful, therefore, in balancing the operating level with the level of demand of the environment so that the full range of transient responsiveness can be attained.

A number of oculomotor systems show adaptive behaviour. For example, the vestibulo-ocular reflex (VOR) is the smooth movement of the eyes which counteracts and thus compensates for rotation of the head. In human subjects after prolonged wearing of reversing prisms, the VOR has been shown to adapt by reversing in direction[1]. In monkeys following extraocular muscle tenectomy, the VOR gain (eye velocity/head velocity) is initially reduced but gradually increases over the course of a few weeks[2]. Saccades, the rapid flick of the eyes that can be observed during reading, normally have an amplitude equal to the target displacement. However, in patients with ocular muscle paresis, the saccadic amplitude is initially reduced, but gradually increases over the course of several days[3]. This is also seen in monkeys with tenectomized ocular muscles[2,4]. In normal subjects, short-term saccadic adaptation can be observed after training with stimuli containing target displacements presented during the saccade[5,6].

Adaptation is also observed following prolonged accommodative effort. The accommodation system detects blur of the retinal image of the target and uses feedback to change lens curvature until the image becomes clear[7]. If the accommodative loop is opened, the lens power decays towards a tonic value which varies from 1 to 4 D depending on the subject[8,9]. Schor examined the effect of different means of opening the accommodative loop on the decay rate[10]. After 5 s of monocular viewing at 2 D, the rapid decay time constants were similar, ranging from 3 to 5 s. However, after 60 s of monocular viewing, the decay rate was much slower and differed according to the means of opening the loop. For example, opening the loop by pinhole viewing, empty field, and darkness for 15 s resulted in decreases of about 0.2, 0.9 and 2.0 D, respectively. Rosenfield and Gilmartin examined the change in lens power after 15, 30 and 45 s near viewing[11]. They found similar decay curves which showed a decrease of about 0.7 D over a 30 s interval. Schor found different accommodative decay rates among individuals with different AC/A and CA/C ratios[10]. (AC/A is the accommodative convergence response to accommodative stimulus ratio. CA/C is the convergence accommodation response to convergence stimulus ratio.)

The vergence system senses disparity between the retinal images in the two eyes and uses feedback to rotate the two eyes until fusion is attained[12]. Normally, if one eye is occluded, thus opening the vergence loop, the angle between the two eyes decays[13–14] towards a tonic value ranging from -1 to $+1$ MA with a time constant[15–17] of ≈ 10 s. However, if binocular fusion is maintained for a longer time (e.g. 1–2 min), the subsequent open-loop decay time constant increases many-fold so that the open-loop vergence level remains close to the binocular

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fusional angle for a number of minutes. This phenomenon has been called prism adaptation[18]. The decay time constants following short and long adaptation periods were generally longer than those for accommodation. Schor reported finding in normal individuals various amounts of decay after blockage of one eye[10,19,20]. For example, in two subjects, after 60 s of binocular viewing at 1 MA, the vergence angle dropped[19,20] $\approx$0.3 MA in 15 s. Also, he found differences in vergence decay time course among normal individuals with different AC/A and CA/C ratios[10].

Experiments involving both accommodation and vergence have also been performed. For example, Schor[16] found in one subject that, after 30 s of binocular viewing at 1 MA and 0.66 D, and subsequent opening of the vergence loop, the vergence angle showed no decrease after 40 s. Wolf *et al.* examined the time course of open-loop accommodation and vergence during and after a period of near vision[21]. The subjects showed an increase in open-loop accommodation output of $\approx$0.3 D after 45 min of reading which declined back to the baseline after $\approx$12 min. Similarly, their open-loop vergence increased by $\approx$0.35 MA and returned to the baseline after $\approx$3 min. Fisher *et al.* examined the effect of stimulus duration, ranging from 0.25 to 8 min, on tonic accommodation and tonic vergence[22]. They found that for tonic accommodation, stimulus duration had no significant effect on the magnitude and rate of decay, whereas for tonic vergence, the after-effect was larger and had a slower rate of decay as stimulus duration increased. Sethi and North used an alternating binocular and monocular viewing paradigm to measure the time course of the phoria value[23]. They presented 15 s of binocular viewing with a prism before the left eye followed by 15 s of monocular viewing (accommodative closed-loop with stimulus at 0.25 D), at which time the phoria was measured. This cycle was repeated 14 times. They plotted their results in terms of the phoria relative to the prism value. The measured heterophoria was initially altered following introduction of the prism. For example, for a 2 MA base-out prism the phoria increased by about 0.67 MA, and for a 4 MA base-out prism, it increased about 1 MA over the course of 13 cycles following the initial cycle. For both cases, however, most of the change occurred during the first 5 cycles.

Several models have been proposed to account for adaptation in the vergence and accommodation systems. Hofmann and Bielschowsky were the first to present a descriptive model of vergence adaptation in which a fast-acting mechanism reduced binocular disparity by convergence of the two eyes and a slow-acting mechanism maintained binocular alignment[24]. In this way, the slow-acting mechanism supported the eye angle and provided comfort during binocular vision. More recently, Wick and London proposed a descriptive model of fusional vergence which included a prism adaptation component[25]. They used the model as a basis for analysing binocular visual function, but did not actually demonstrate the adaptation process.

Schor developed the first quantitative model for simulating the vergence adaptation process[19]. The model consisted of a fast neural integrator of the form

$$\frac{k_f}{s + a_f}$$

and a slow neural integrator of the form

$$\frac{k_s}{s + a_s}.$$

The step response of the fast neural integrator, for example, would be a rising exponential with a time constant of $1/a_f$ and a final level of k_f/a_f. A similar description holds for the slow neural integrator. The slow neural integrator is summed with a parallel unity gain element to result in an equivalent element of the form

$$1 + \frac{k_s}{s + a_s}.$$

The fast neural integrator and the equivalent element are placed in series in the forward loop of the model. However, it should be noted that the model lacked a true tonic component so that if the feedback loop is opened, the model response would go to zero rather than the tonic value.

Neglecting the static non-linearities, time delay, and rapid plant dynamics elements, the overall transfer function of the closed-loop system is given by

$$\frac{k_f[s + (a_s + k_s)]}{s^2 + (a_f + a_s + k_f)s + [a_f a_s + k_f(a_s + k_s)]} \qquad (1)$$

Since a_f, a_s, k_f and k_s are constant parameters, *Equation 1* is a linear second-order transfer function. However, this form of transfer function exhibits no adaptive properties. That is, the dynamics of the response is the same regardless of the past history of the stimulus. It is unlikely that the static non-linearities such as threshold and saturation can account for the experimental adaptation that occurs over a wide range of stimulus levels. Therefore, it is not surprising that Schor showed only simulations of closed-loop dynamics but no adaptation effects in his model[19].

More recently[10], Schor expanded on the earlier model[19] to include the accommodative feedback loop and the cross-link components accommodative convergence (AC) and convergence accommodation (CA). Rosenfield and Gilmartin[26,27] proposed a similar model except they removed the fast components. Also, based on their experimental results, indicated that cross-link interactions do not directly stimulate oculomotor adaptation, they placed the cross-link after the adaptive component in each loop. It can be shown, however, that both of these models can be reduced to overall linear transfer functions similar to *Equation 1*. Therefore, these models also do not exhibit adaptation properties even though some of the elements were called adaptive components. Indeed, no simulations of the adaptation process were presented for either model.

The model proposed below was based on an earlier successful static dual-interactive model of the accommodation and vergence system. The approach to the development of the model was to introduce the fewest number of components that will give responses consistent with the experimental findings. In addition, the model was configured so that when it returns to the steady state condition over time, it becomes identical to the original static model.

Methods

Model configuration

The components of the basic accommodation and vergence systems are similar (*Figure 1*). Both contain a controller in the forward-loop. The controller consists of a transient and a sustained component. The transient component drives the initial dynamic portion of the response, whereas the sustained component maintains closed-loop feedback of the steady state level[12,28–30]. The tonic component represents the value of accommodation or vergence when the feedback loop is opened[14,31]. This is done by opening the switch in the model. In this way, the controller output goes to zero, so that eventually only the tonic component drives the response.

To account for the change in decay characteristics following sustained fixation, an adaptive component has been incorporated into the basic model (*Figure 2*). The time delays of the vergence (≈ 200 ms)[12] and accommodation (≈ 350 ms)[7] systems were neglected in the model for two reasons. First, the time delays are much shorter than the adaptation decay time constants. Thus, the large time scale for plotting the results will not show the delays. Second, it has been shown that for stability the early portion of the initial response consists of an open-loop movement[29,30], which is followed by a slow closed-loop movement that accurately tracks the target. Therefore, neglecting the time delays will not affect the overall dynamic characteristics of the adaptation model. Also, for simplicity, the transient element was deleted since its response is completed quickly, whereas adaptation takes place over a much longer time interval.

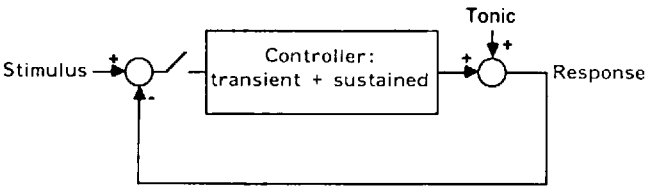

Figure 1 General non-adaptation model of accommodation or vergence. With the switch closed, the error, or stimulus minus response, drives the controller. The controller in the forward loop consists of transient and sustained components. The output of the controller is summed with the tonic component to give the response. To simulate the open-loop condition, the switch is opened, and the controller output goes to zero, so that the response equals the tonic value

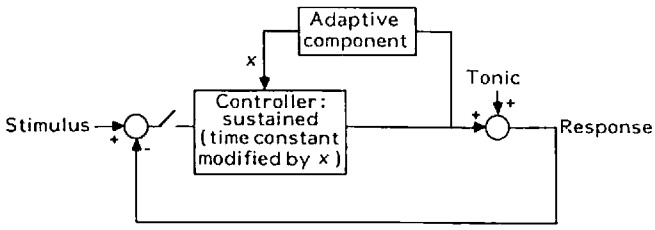

Figure 2 General adaptation model of accommodation or vergence. For simplicity, the transient element in the controller is removed. The configuration is basically the same as the general non-adaptation model, except the output of the controller also drives the adaptive component. The adaptive component output, x, modifies the time constant of the sustained element in the controller. An increase in stimulus level increases the controller output, which in turn increases x, and thereby the controller time constant. If the switch is opened to simulate the open-loop condition, the controller decays toward zero, but at a much slower rate because of the increased time constant. Eventually the response returns to the tonic value

The adaptive component receives its input from the output of the controller. This is consistent with the concept that adaptation is related to the effort of vergence or accommodation[20]. The unique feature of the model is that the sustained element time constant is modified by the adaptive component output. Although this configuration is unique among near response oculomotor models, the modification of a component's time constant is seen in other systems. For example, in the saccadic system, Optican and Miles[4] simulated adaptation using modification of time constants. Also, it has been shown that for blood glucose regulation the fractional loss per hour of glucose varies with the total insulin level[32]. Thus, in the accommodation and vergence systems, as controller output increases with increased effort, the adaptive component increases the time constant of the sustained element. If at that point the feedback loop is opened, the decay of the sustained element will take longer, thereby simulating the longer decay time following prolonged adaptation found experimentally.

The combined accommodation and vergence adaptation model is shown in *Figure 3*. Consider first the accommodative loop. The deadspace element (*DE*) represents the depth of field. The controller output is multiplied by factor m_A and input to a tanh function which serves as a comparison element (CE). The factor m_A is used to provide an appropriate operating range on the abscissa of the tanh function. The compression element reduces the controller output for large magnitude inputs so that the adaptation effect is not drastically different at various adapting stimulus levels. The adaptive component is represented by the first-order dynamic element

$$\frac{1}{T_{A1}s + 1},$$

where T_{A1} is the accommodative adaptation time constant. The accommodative adaptation gain, K_A, controls the magnitude of the adaptive component output level. The time constant of the accommodative sustained element, T_{A2}, is added to the absolute value of the cube of the adaptive component output, $|a^3|$, to give the controller time constant. The cubic relationship was obtained empirically to provide negligible increase in time constant for a small amount of adaptation, but very long time constant for a large amount of adaptation. The deadspace operator represents the depth of field (for simplicity assumed constant at ± 0.25 D). The accommodative controller gain, *ACG*, is the static gain of the accommodation system[14].

A similar configuration applies to the vergence system, where the deadspace element (*DE*) represents Panum's fusional area[34] (for simplicity, assumed constant at ± 15 min arc or ± 0.07 MA). Also, for the overall model, the cross-link elements *AC* and *CA* represent accommodative convergence and convergence accommodation, respectively. In addition, *ABIAS* and *VBIAS* represent tonic accommodation and tonic vergence, respectively.

An important feature of this new model is that the static conditions should be identical to an earlier successful static interactive dual-feedback model[14]. Indeed, using the final value theorem and setting s to zero, the controller reduces to gains *ACG* and *VCG*, and the model becomes identical to the previous static model.

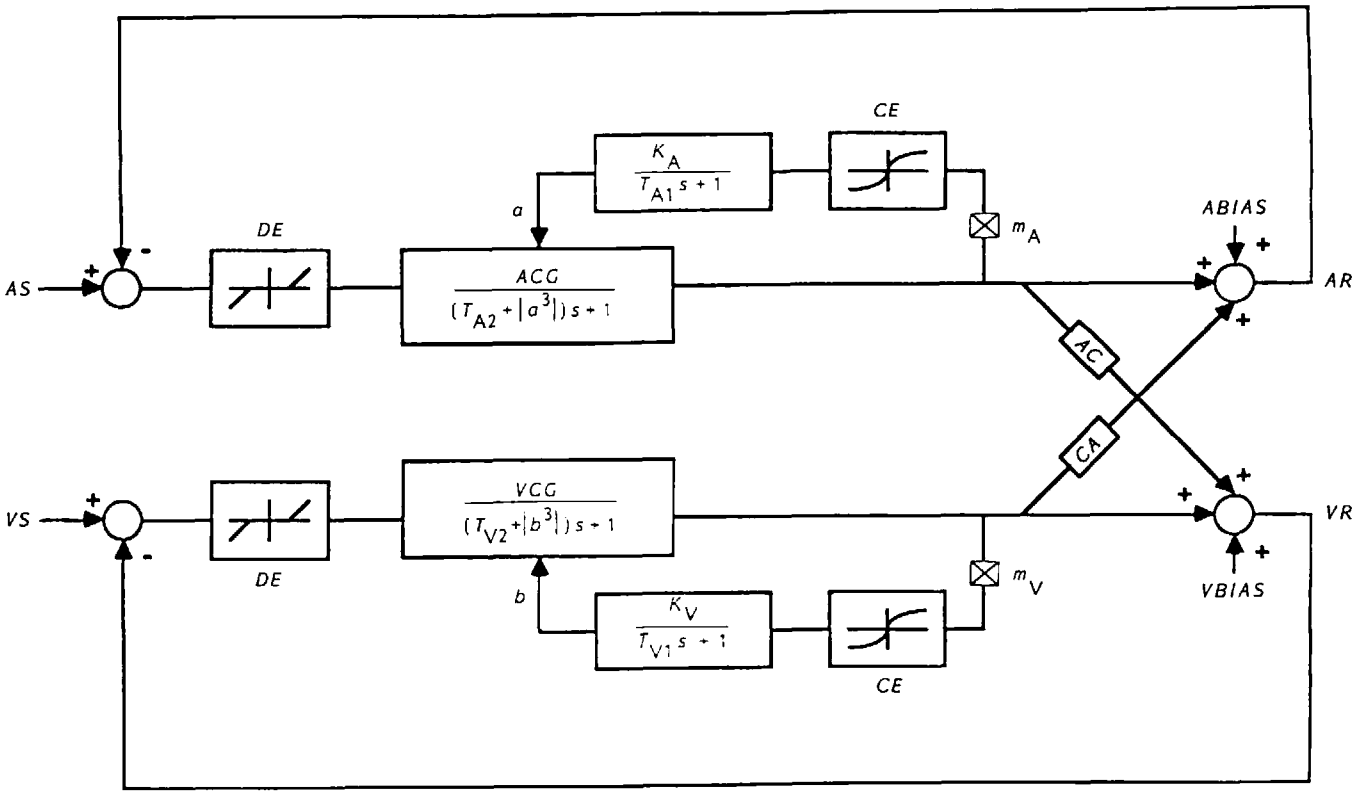

Figure 3 Adaptation model of combined accommodation and vergence system. See text for details. The accommodative and vergence controller outputs multiply the factors m_A and m_V, respectively, before input to tanh functions to the compression elements (CE) consisting of tanh functions, which serve to reduce the effect of large amplitude controller values

Model simulation

A program written in Quick-BASIC was used to simulate the model in *Figure 3*. A fourth-order Runge–Khutta integration routine was used to compute the outputs of the linear dynamic elements. The model was simulated on an IBM-PC with EGA board, and the results were plotted with a Laserjet plotter.

Simulations were performed using parameter values from four different subjects obtained previously[14] (see first six parameters in *Table 1*). The parameter values for fit to the averaged experimental data are shown in *Table 1* (last eight parameters). The time constants T_{A2} and T_{V2} were estimated initially from data obtained in blocking or darkness experiments[10,20,35]. On the other hand, the initial values for parameters K_A, K_V, T_{A1} and T_{V1} were estimated from expected dynamics[29,30]. A trial and error procedure was used to obtain values for the gains and time constants. The accommodation system was simulated first, and the parameters were varied until the decay time constants fell within 20% of experimental values. This was similarly done for the vergence system. Then the interactive accommodation and vergence system was used to simulate the Sethi and North experimental results[23]. The parameters were adjusted again so that the magnitudes and time constant were within 10% of the experimental values. These new parameter values were also used to simulate the isolated accommodation and vergence system responses. This procedure was repeated until magnitudes and time constants were within 10% of average experimental values for both isolated and combined conditions.

Table 1 Model parameters for four subjects

	Subjects			
Parameter	*J.S.*	*G.H.*	*F.R.*	*J.M.*
ABIAS (D)	0.42	1.70	0.75	2.20
VBIAS (MA)	−0.62	1.00	0.19	0.93
ACG	8.53	2.60	10.70	2.32
VCG	293.0	141.0	126.0	166.0
AC (MA D^{-1})	1.15	1.76	0.49	0.63
CA (D MA^{-1})	0.66	0.50	0.85	0.47
K_A	4	10	12	10
T_{A1} (s)	4	4	4	4
T_{A2} (s)	25	25	25	25
m_a	3	0.5	0.5	0.5
K_V	9	6	6	5
T_{V1}	8	8	8	8
T_{V2}	50	50	50	50
m_V	0.5	2	2	2

The first six parameters were obtained from reference 14. The last eight parameters were obtained from simulation for fit to averaged experimental data.

Simulation results

As the results for the four subjects were similar, for simplicity only those for one subject, J.S., will be discussed in detail below. The model gains and time constants are shown in *Table 1*. It was found that the desired large amplitude saturation effect was obtained when the input to the tanh function was multiplied by $m_A = 3$ in the accommodation system and by $m_V = 0.5$ in the vergence system (see *Table 1*).

The accommodation-only response was simulated with the vergence feedback open-looped. The accommodative open-loop decay is shown after fixation for 5 and 120 s (*Figure 4*). The accommodative controller time constant is shown as the dashed curve. Note the relatively rapid

decay (time constant of ≈ 10 s) of fixation as contrasted with the slow decay (decrease of ≈0.5 D over a 50 s interval) after 120 s of fixation. Similarly, the vergence-only response was simulated with the accommodation system open-looped. The vergence decay after being open-looped following 5 and 120 s binocular fixation and the vergence controller time constant are shown in *Figure 5*. As above, there was a rapid decay following the shorter fixation interval (time constant ≈ 15 s), but a much slower decay following the longer interval fixation (decrease of ≈0.3 MA over 50 s).

The model was also used to simulate the results of Sethi and North[23] by means of the paradigm described by Henson and North[18]. Initially, both feedback loops were closed with the accommodative stimulus equal to 0.25 D. After 15 s, the accommodative loop as opened for another 15 s. The cycle was then repeated. The simulation results for closed-loop vergence stimulus equal to 2 and 4 MA are shown in *Figure 6*. Note that the bottom of the decay curve, where presumably the phoria measurement would be made, increases towards the vergence stimulus level over time. The amount of change of the phoria value starting from the first cycle is about 0.7 and 1.2 MA for the 2 and 4 MA stimuli respectively.

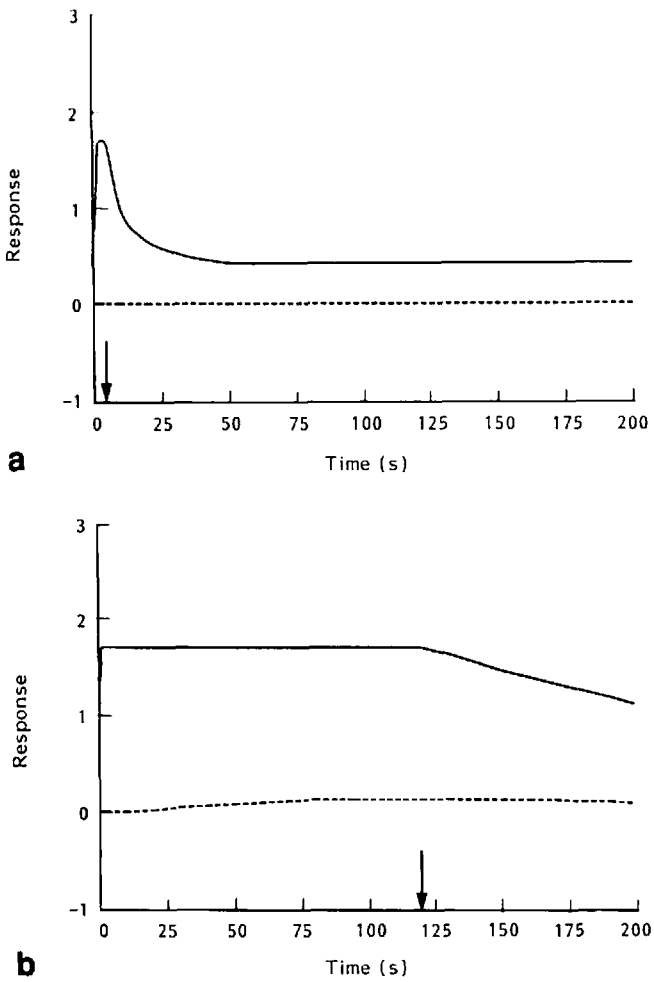

Figure 4 Model simulation of accommodative output (—, D) and controller time constant (---, × 1000 s). The induction stimulus = 2 D. The switch is opened at 5 s (*a*) and 120 s (*b*) to simulate open-loop accommodation. Note the rapid decay following the short induction interval and the much slower decay following the long induction interval. Parameter values for subject J.S. See reference 10 for comparison with experimental curves

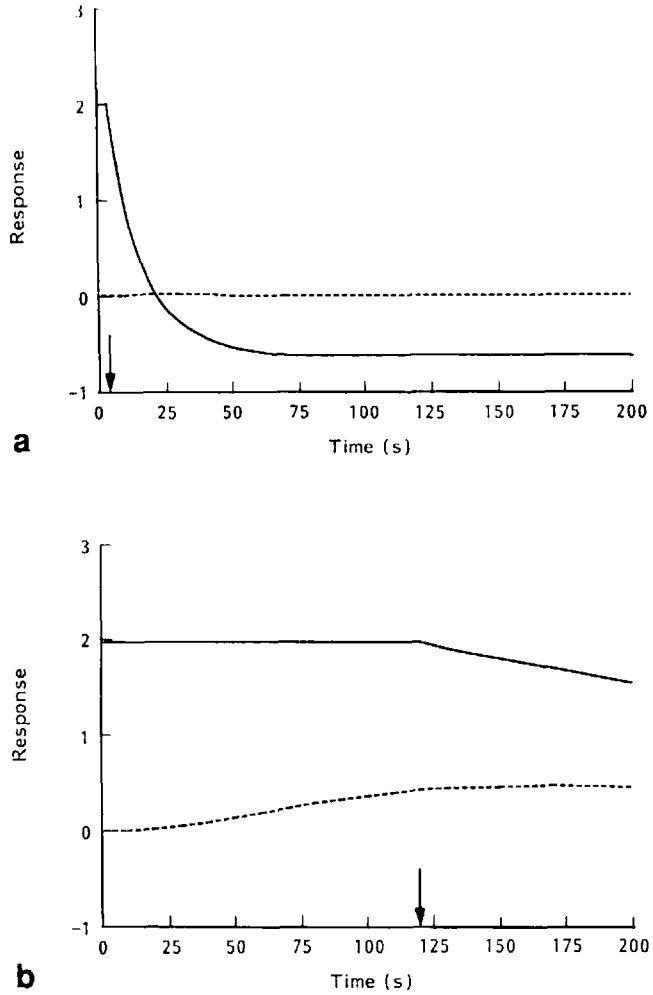

Figure 5 Model simulation of vergence output (—, MA) and controller time constant (---, × 1000 s). The induction stimulus = 2 MA. Switching conditions are the same as in *Figure 4*. Parameters for subject J.S. See reference 10 for comparison with experimental curves

The effect of different amounts of adaptation was also simulated using the Henson and North[18] paradigm. The vergence responses for normal, abnormal, and non-adaptive conditions are shown in *Figure 7a*. Note that the bottom of the decay curve increases toward the vergence stimulus level over time for the normal, to a lesser extent for the abnormal, and not at all in the case of no adaptation. The amount of change of the phoria value starting from the first cycle is ≈1.07, 0.50 and 0 MA, for the normal, abnormal, and non-adaptative conditions, respectively. The time course of the controller time constants are shown in *Figure 7b*. In the normal condition, the time constant continues to rise over time, whereas for the abnormal condition, the time constant reaches a saturation level and thereafter remains at the same level. Saturation of the time constant occurs at an even lower level for the non-adaptive condition.

To investigate the effect of other changes in stimulus on the adaptive component output, the vergence stimulus was decreased from 4 to 1 MA in 1 MA steps while accommodation was open-looped. The vergence response (*Figure 8*) followed the stimulus accurately. The vergence controller time constant is seen to increase and then decrease as the stimulus level decreases, eventually returning to the initial value.

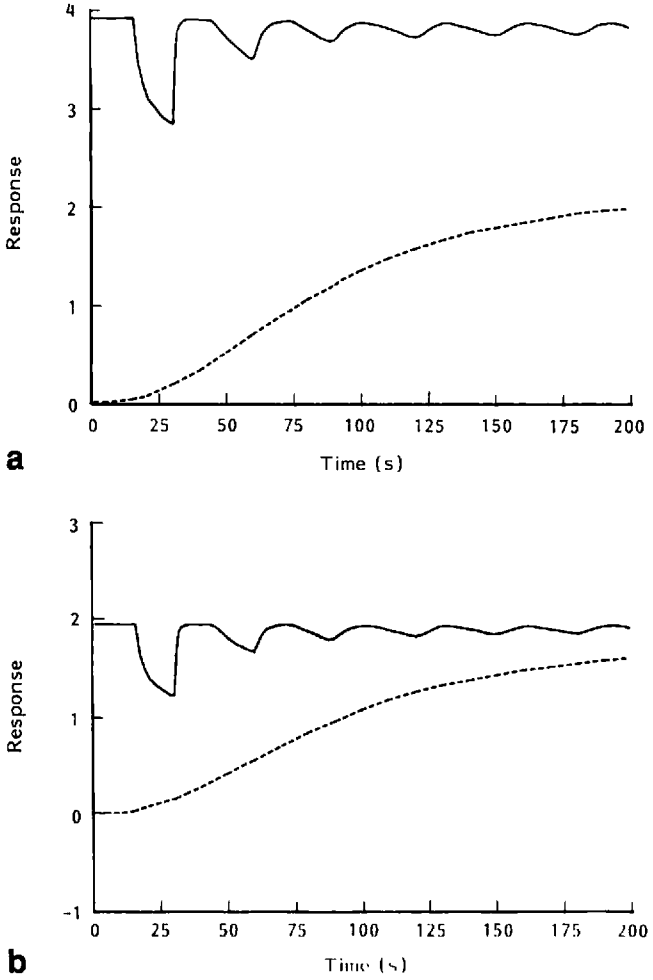

Figure 6 Model simulation of data from the Sethi and North[23] experiment in which vergence loop alternates between 15 s closed-loop and 15 s open-loop. Accommodative loop is closed with accommodative stimulus equal to 0.25 D. Vergence output (—, MA) and controller time constant (---, × 1000 s) are shown for closed-loop vergence stimulus of (*a*) 4 MA and (*b*) 2 MA. For each cycle, the vergence output after the vergence loop is opened for 15 s represents the phoria value. Note the greater amplitude of change in the phoria for the large as compared to small amplitude stimulus. Parameters for subject J.S.

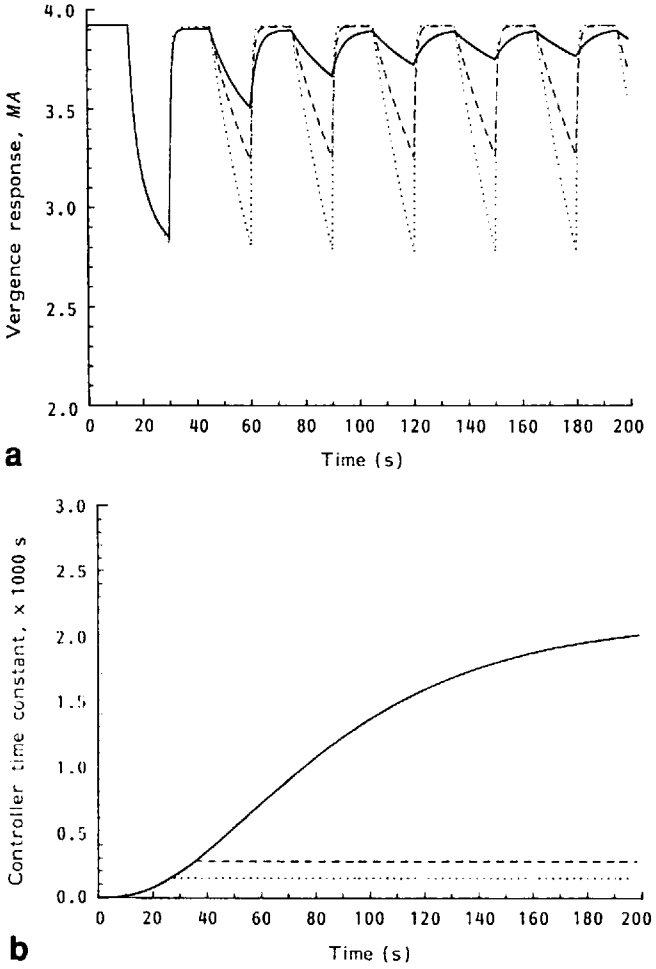

Figure 7 Model simulation of different amounts of adaptation using the Henson and North[18] paradigm. (*a*) Vergence responses are shown for the normal (—), abnormal (---), and non-adaptive (•••) conditions. (*b*) Controller time constants for the three conditions. Note that the saturation level for the controller time constant is large for the normal, intermediate for the abnormal, and small for the non-adaptive condition

Discussion

Several models have been proposed previously[10,16,19,20,25,27] to describe the accommodative and vergence adaptation behaviour. However, no simulations of these models have been presented to show the adaptation process quantitatively. Indeed, analysis of these models indicates that they would not be capable of demonstrating any adaptation to induction stimuli. The present model is not only able to show adaptation to accommodative and vergence stimuli, but also exhibits the useful property that it returns identically to the previous model of accommodation and vergence under static conditions[14]. Therefore, this is the first quantitative model which simulates accurately accommodative and vergence behaviour under static as well as dynamic adaptive conditions.

The relatively rapid open-loop decay of accommodation and vergence before induction reflects the small accommodative and vergence stimulus magnitudes normally present. Indeed, for far viewing, the adaptive

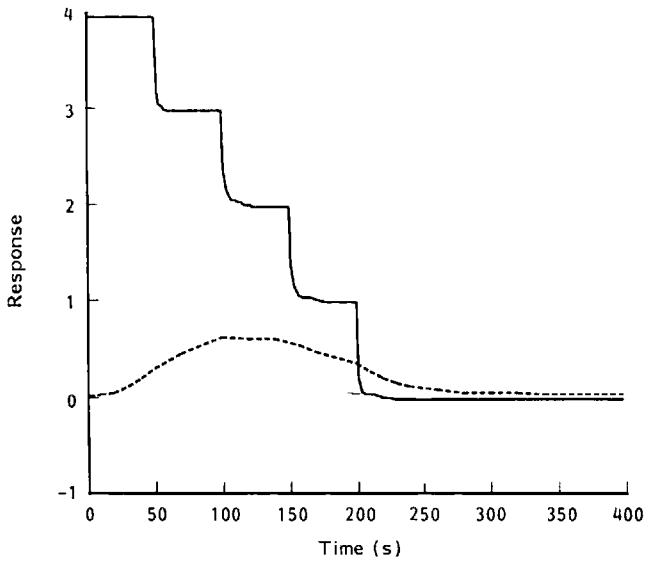

Figure 8 Model simulation of closed-loop vergence response (—, MA) and controller time constant (---, × 1000 s) for decreasing steps of stimuli starting at 4 MA. Note the increase, leveling off, and gradual decline in the time constant value as the stimulus level decreases. For stimulus level of 0 MA, the controller time constant returns to its initial small value. Note the time range is 400 s. Parameters for subject J.S.

element output is quite small and adds minimally to the sustained element time constant. Therefore, the model should show a reduction in adaptive element contribution as stimulus level is decreased. *Figure 8* shows the closed-loop vergence response and controller time constant with decreasing steps of vergence stimulus. The controller time constant rises for the 4 MA stimulus and levels off as the stimulus decreases, finally declining and reaching nearly zero contribution for 0 MA stimulus. At this point, the controller time constant returns to the original rapid rate seen before induction.

In the simulation of Sethi and North[23] data (*Figure 6*), the rising portion of the vergence response at the beginning of each cycle became more rounded in shape as adaptation progressed. However, the closed-loop response shape would be much sharper if a transient portion were added to the controller. Hung *et al.*[29,30] proposed that the accommodation and vergence systems exhibit dual-mode behaviour, in which the transient component responds with an open-loop movement based on the estimate of error and error rate. This is then followed by a closed-loop sustained response which provides slow adjustment for the residual error. Thus, incorporation of the transient component would give rapid closed-loop responses but retain the open-loop decay seen in the experimental results.

The model simulation decay rates were close to those found experimentally. However, they were not exactly equal to experimental values because there were a number of different, sometimes conflicting, demands within the model. Indeed, most of the parameter values obtained were compromise solutions. For example, a large K_V increased the decay amplitude and simulated more closely the data of Sethi and North[23]. However, this decreased the decay rate in the simulation of blockage after 120 s of binocular viewing to a rate slower than that found experimentally[16,19]. Thus, a compromise value of K_V gave reasonable, but not exact, simulation of the experimental time course curves.

Rosenfield and Gilmartin[27] found that during 3 min of binocular near vision with accommodation, open-looped convergence-accommodation was not significantly altered whereas heterophoria exhibited a change, indicating adaptation in the vergence system. They therefore concluded that both reflex (corresponding to the transient and sustained components in the present model) and adaptive vergence contribute to *CA* in such a way that *CA* remained constant. That is, the cross-link followed the reflex and adaptive components. However, as mentioned in the Introduction, their model actually was not capable of simulating adaptation. Also, they did not discuss how the adaptive component would automatically compensate for changes in the reflex component to result in a constant *CA*.

Nevertheless, the model presented here is consistent with the results of Rosenfield and Gilmartin. Its configuration offers a means of reconciling the constancy of *CA* while vergence adaptation takes place. Importantly, it must be recognized that *CA* is measured under closed-loop vergence and open-loop accommodation, whereas heterophoria is measured under closed-loop accommodation and open-loop vergence. The model *CA* cross-link occurs after the vergence controller. Thus, when the vergence feedback loop is closed, the controller output drives both the vergence response, and via the *CA* cross-link, the open-loop accommodative response. Since the controller maintains the vergence error to within 30 min arc, it must remain relatively constant under static closed-loop conditions[14]. Therefore, *CA* also remains relatively constant. This constancy of *CA* during closed-loop vergence has also been demonstrated in simulations of the model (not shown).

On the other hand, during a heterophoria measurement, the vergence loop is opened, and the vergence controller output decays with a time constant determined by the output of the adaptive component. If the time constant is long, corresponding to a long period ($\geqslant 1-2$ min) of binocular viewing, the controller output will remain near its closed-loop value for some time. The change in vergence output, or the heterophoria, will reflect the increase in controller output, thus indicating vergence adaptation. Hence the present model is consistent with the results of Rosenfield and Gilmartin[27] and reconciles the difference between the *CA* measure and adaptation of the heterophoria.

Schor and Horner[36] found a greater amplitude and duration of accommodative after-effect in convergence insufficiency than convergence excess subjects. Also, they found for vergence after-effect a reverse trend for the two groups. During the course of simulations of the present model, it was found that the adaptive gains, K_A and K_V, were sensitive to static controller gains *ACG* and *VCG* (not shown in the results). Thus, a subject with a low *ACG* and high *VCG* may be counteracted by high K_A and low K_V so that during a period of adaptation, the accommodative controller moves more rapidly to its static gain value than the vergence controller. The net effect may be a normalization tendency so that the symptoms are less apparent.

The data for the four subjects in *Table 1* provide an assessment of the variability of parameters among subjects. The time constants T_{A1}, T_{A2}, T_{V1} and T_{V2} were the same for all subjects. The soft saturation input multiplication factors, m_A and m_V, were the same in three of the subjects. However, the gains of the adaptive components, K_A and K_V, were different for the four subjects. Thus, the adaptive component gains provided the fine adjustments for accurate simulation of experimental results. This also suggests that the adaptative component may help to compensate for controller gain values that may differ widely among various individuals in the normal population.

Future developments of the model include detailed sensitivity analysis of controller and adaptive components to parameter variations, simulation of adaptive behaviour in symptomatic individuals, and application to the clinical evaluation of prism adaptation. In addition, inclusion of transient elements and proximal components, along with stimulus boundaries for error and error velocity, will result in a comprehensive model of the interactive dual-feedback model of accommodation and vergence. Finally, this model may serve as the foundation for quantitative analysis of new experimental findings of accommodative and vergence behaviour.

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